

Portfolio

20 Feb, 2026

Personal portfolio

Virgi Atha Raditya

virgiatharaditya37@gmail.com

 [Virgi Atha Raditya](#)

+62 895 0712 6276



hello, i'm Virgi Atha Raditya!

I am a Computer Science graduate with a strong interest in data analysis and artificial intelligence development. I have experience analyzing data, building and deploying AI models for prediction tasks, and developing data-driven solutions. I have also developed web applications using CodeIgniter 4 and Laravel and gained hands-on experience through internships that strengthened both my technical and collaboration skills. I enjoy transforming data into meaningful insights, learning new technologies, solving complex problems, and contributing to impactful projects in data analytics, AI, and intelligent systems.

personal skills

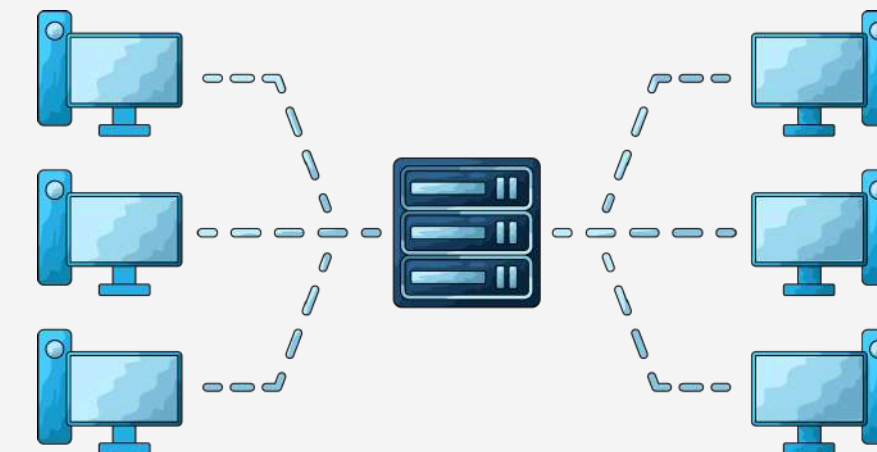
Computer Science student interested in web development and software engineering, experienced in building user-friendly applications and working collaboratively in teams.



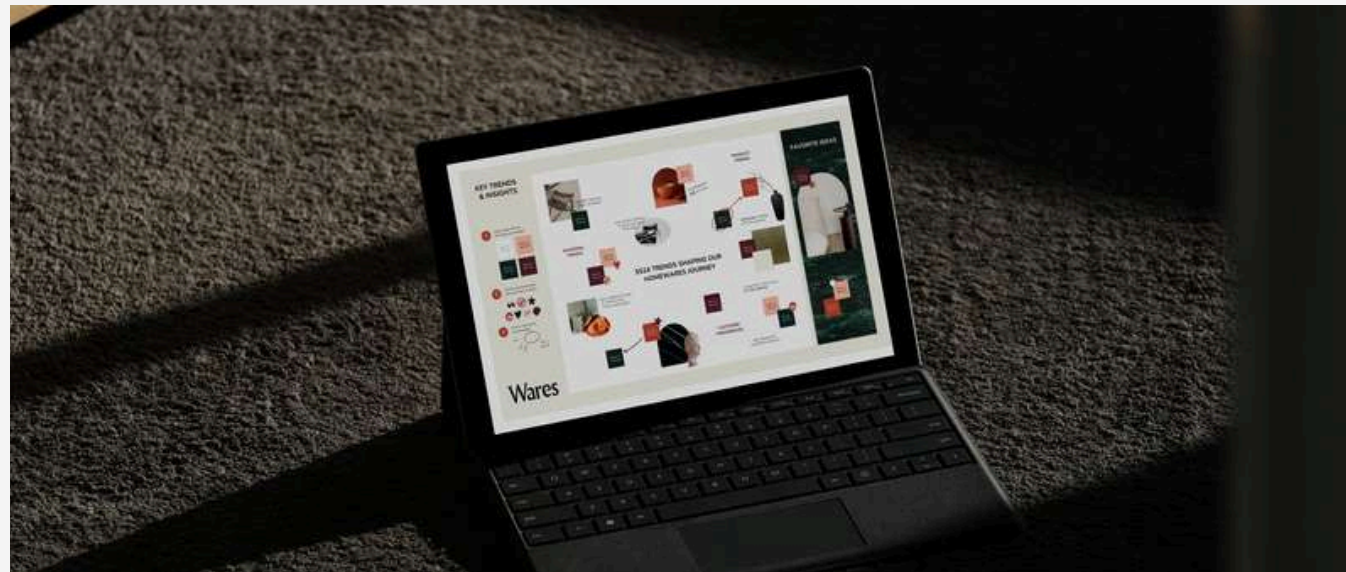
Data Analysis



Web Development



Internet of Things



my work experience

Experienced in developing web-based systems, assisting network engineering practices, and supporting IT infrastructure across different environments.

Full Stack Developer, Stasiun Meteorologi Syamsudin Noor in Banjarmasin

Developed siBisa, a web-based inventory and internal loan management system using CodeIgniter 4 with QR Code API and Telegram Bot integration.

Teaching Assistant Network Engineer Course

Assisted LAN installation, network topology design, and guided configuration using Cisco Packet Tracer and Debian web server deployment.

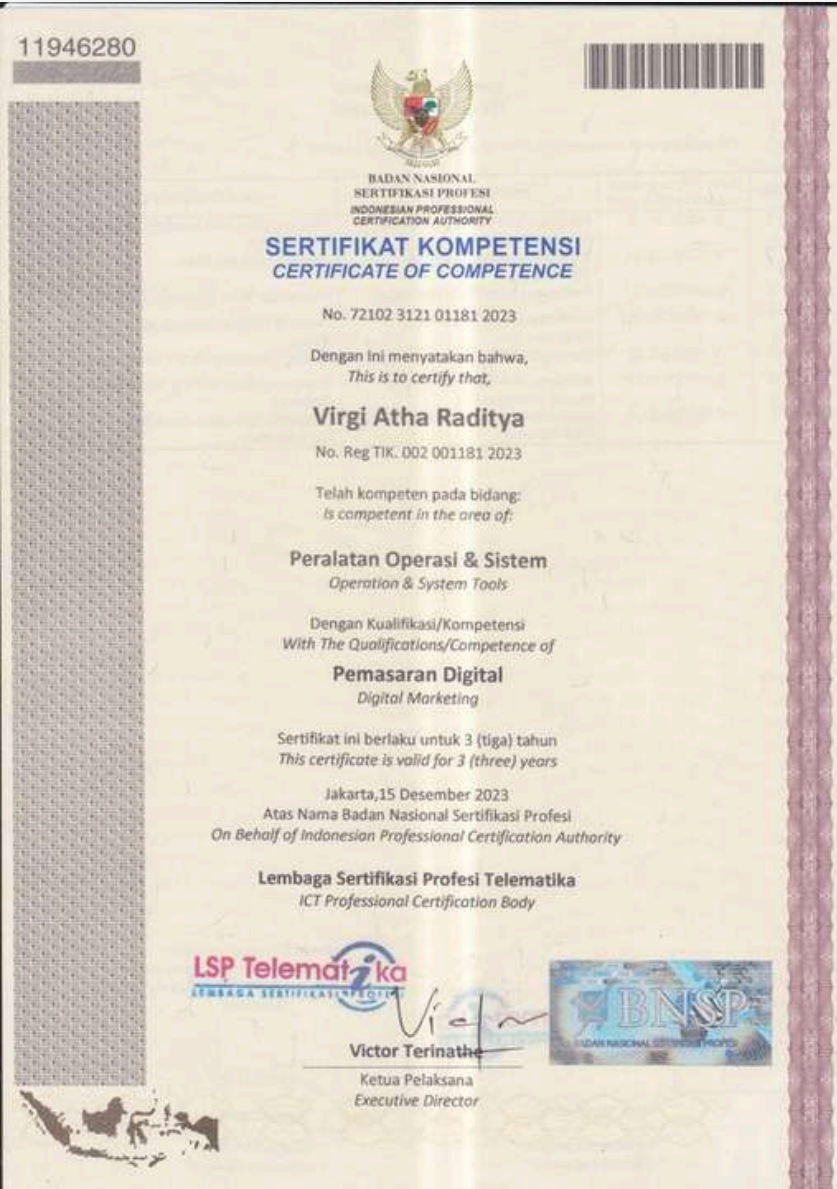
Teaching Assistant Numerical Analysis course

Supported the delivery of the Numerical Analysis course by assisting lecturers and guiding students in applying numerical methods to computational problems.

IT Support Technician, PT. Angkasa Pura I

Learned and supported airport IT and communication systems and developed a VoIP communication system.

Certifications



Digital Marketing - BNSP



Junior Computer Operator
- BNSP



Microsoft Office 2019
Skills Program - LKP Duta

Personal Presentation

20 Feb, 2026



MSIB Batch 7 participants with the AI 4 Jobs program

Virgi Atha Raditya

virgiatharaditya37@gmail.com

[Virgi Atha Raditya](#)

+62 895 0712 6276

Projects

Jurnal Teknik Informatika (JUTIF) Vol. 7, No. 1, February 2026, Page. 633-646
P-ISSN: 2723-3863 <https://jutif.if.unsoed.ac.id>
E-ISSN: 2723-3871 DOI: <https://doi.org/10.52436/1.jutif.2026.7.1.5424>

Analysis of Static and Contextual Word Embeddings in Capsule Network for Sentiment Analysis of The Free Nutritious Meal Program on Twitter

Virgi Atha Raditya¹, Triando Hamonangan Saragih^{*2}, Mohammad Reza Faisal³, Friska Abadi⁴, Muliadi⁵

^{1,2,3,4,5}Department of Computer Science, Lambung Mangkurat University, Banjarbaru, Indonesia

Email: ²triando.saragih@ulm.ac.id

Received : Nov 4, 2025; Revised : Nov 14, 2025; Accepted : Nov 17, 2025; Published : Feb 15, 2026

Abstract

Public discourse surrounding Indonesia's Makan Bergizi Gratis (MBG) program reflects diverse opinions that have not yet been systematically examined using computational methods. This study addresses that gap by evaluating the effectiveness of static and contextual word embeddings within a Capsule Network (CapsNet) framework for sentiment analysis of MBG-related tweets on Twitter. A total of 7,133 Indonesian-language tweets were collected through web crawling, preprocessed, and manually labeled into positive, neutral, and negative categories. Four embedding techniques—Word2Vec, FastText, ELMo, and IndoBERT—were tested under two preprocessing settings, raw and stemming. The experimental results show that Word2Vec on raw text achieved the highest accuracy of 96.17%, while FastText obtained the best performance on stemmed data with 94.10%. These findings indicate that morphological normalization benefits static and subword-based embeddings, whereas contextual models maintain stable performance without extensive fine-tuning. Overall, this study demonstrates the potential of combining CapsNet with appropriate embedding strategies for Indonesian-language sentiment analysis and provides evidence that natural language processing can support data-driven evaluation of public programs such as MBG.

Project 1: Sentiment Analysis Project

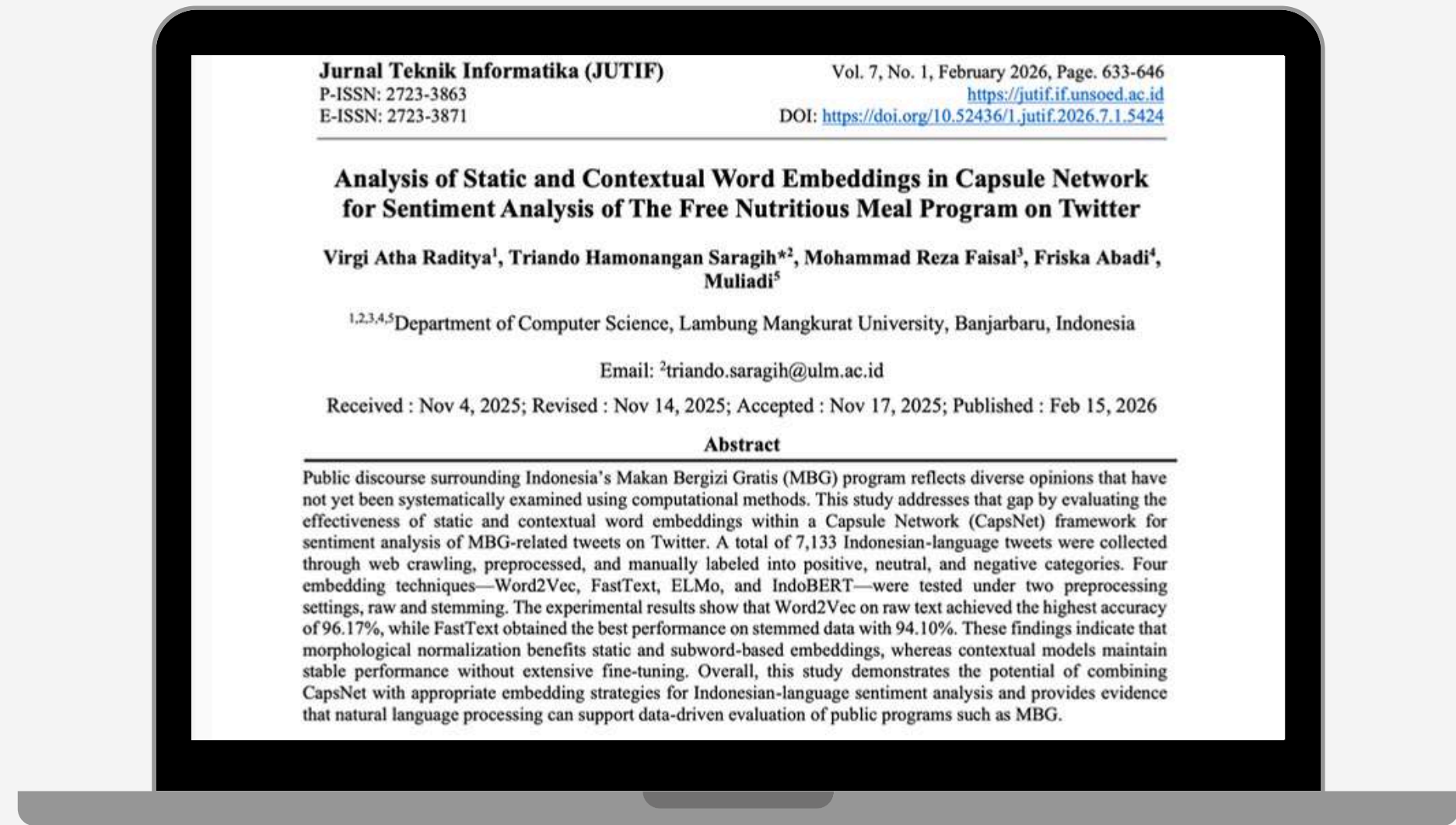
Sentiment Analysis Project

Analyzing Public Sentiment Toward the Makan
Bergizi Gratis (MBG) Program Using Twitter Data

Published in SINTA 2 Journal (JUTIF)

Project Overview

This project analyzes public sentiment toward Indonesia's Makan Bergizi Gratis (MBG) program using Twitter data. The goal is to classify opinions into positive, neutral, and negative sentiments using NLP techniques.



Problem

Public reactions to the Free Nutritious Meal (MBG) program are widely expressed on social media, but analyzing thousands of tweets manually is inefficient and time-consuming. Without systematic analysis, it is difficult to understand the overall public sentiment toward the program.

Solution

This project applies sentiment analysis using Natural Language Processing (NLP) to automatically classify Twitter posts into positive, neutral, and negative sentiments, enabling a data-driven understanding of public opinion toward the MBG program.

Ref	Method	Accuracy
[23]	CapsNet	96%
[24]	BiLSTM + CapsNet + GloVe	98.80%
[25]	BiLSTM + ELMo	51.33%
[26]	LSTM + FastText	89.11%
[27]	BiLSTM + FastText	89.80%
Our Proposed Method	CapsNet + Word2Vec (Stemming)	96.17%

Table 1. Comparison with Related Works

Key Insights

- WordWord2Vec achieved the highest accuracy when the text retained its original morphology.
- FastText performs better when words are normalized because it relies on subword structures.
- ELMo and IndoBERT provide consistent results but require higher computational resources.

Business / Policy Implications

The results demonstrate that social media sentiment analysis can support data-driven policy evaluation.

- Potential applications:
- Monitoring public response to government programs
- Detecting misinformation trends
- Evaluating public satisfaction toward policies

This approach enables policymakers to make more evidence-based decisions using large-scale social media data.

Tools & Technologies Used

Programming & Analysis

- Python
- Google Colab

Libraries

- Scikit-learn
- TensorFlow
- Gensim
- FastText
- Matplotlib
- Seaborn

Data Processing

- NLP preprocessing
- sentiment labeling
- machine learning modeling

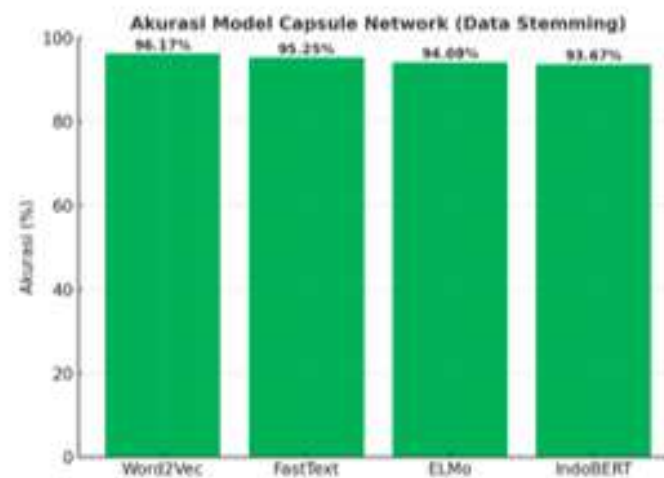


Figure 4. Comparison of Model Accuracy on Stemming Data

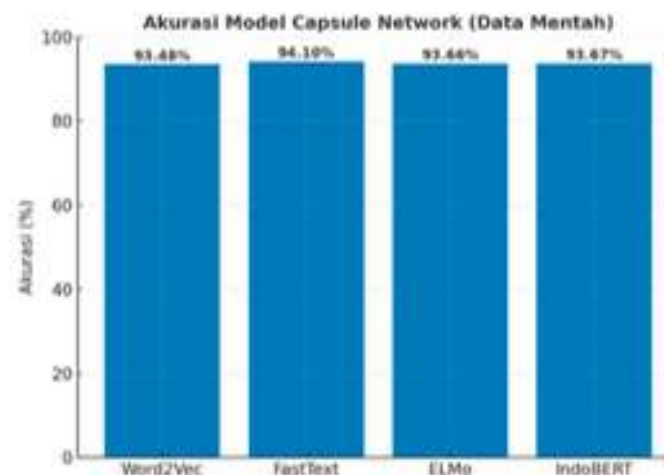
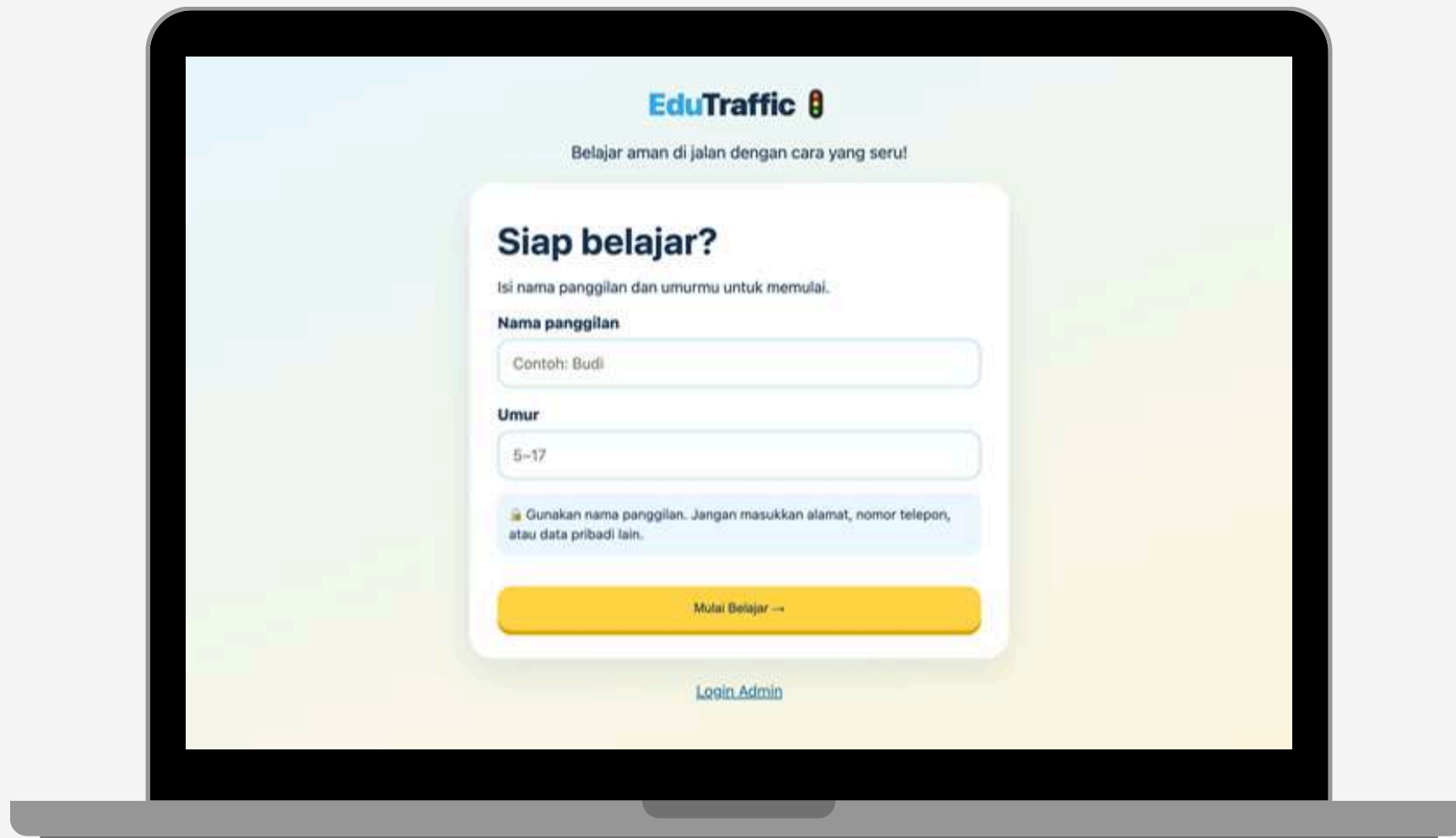


Figure 5. Comparison of Model Accuracy on Raw Data

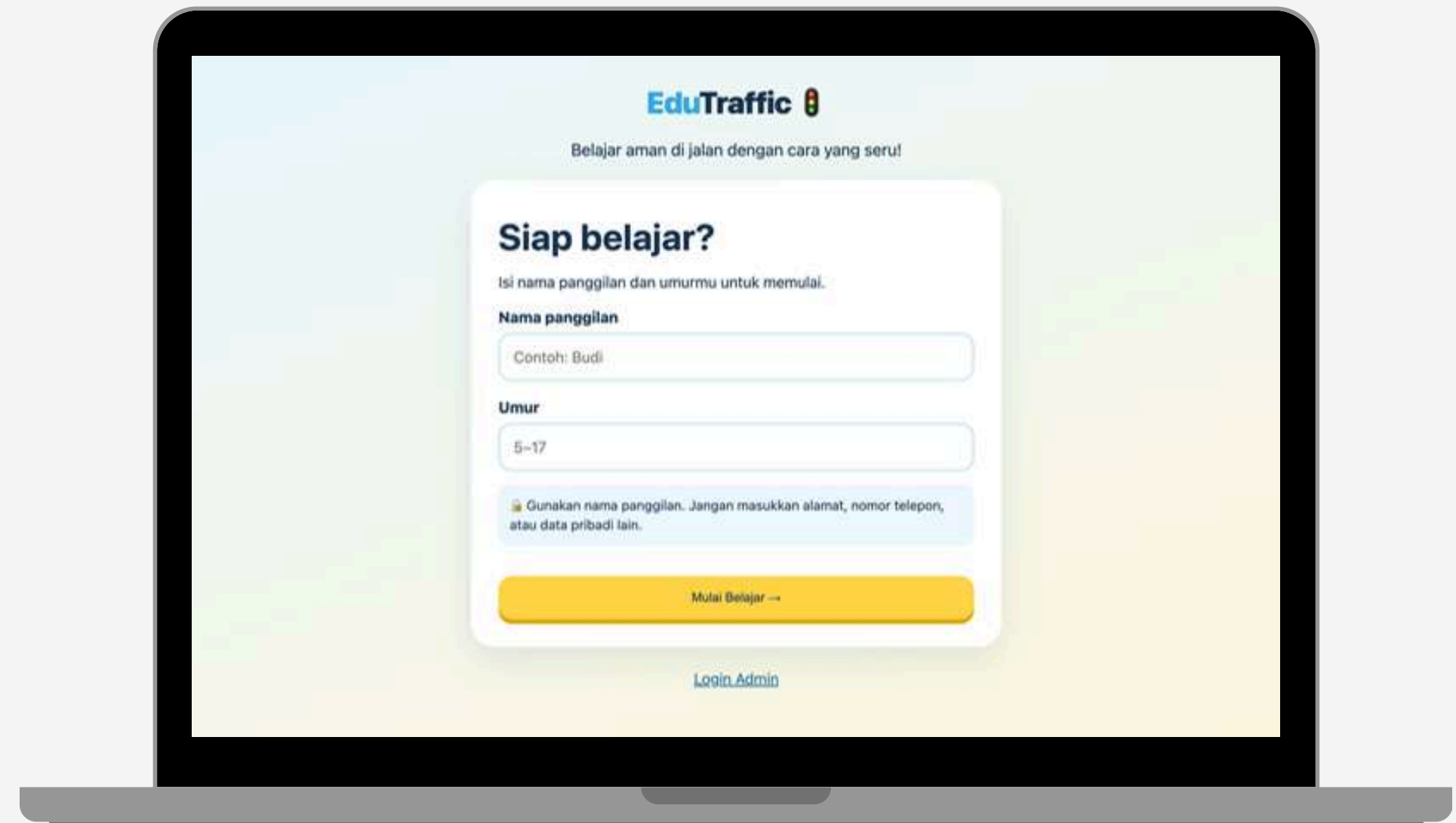


Project 2: EduTraffic: Education of Traffic

Web Development Project

Project Overview

Developed EduTraffic, a child-friendly web-based traffic safety education platform that combines interactive quizzes with personalized learning feedback. The system evaluates participants' knowledge of driving and pedestrian safety, provides detailed results and AI-generated conclusions, and features an admin dashboard for managing questions and monitoring participant performance.



Problem

Traffic safety education for children is often delivered through static materials that are less engaging and provide limited insight into each child's understanding. Without an interactive assessment system, it is difficult to identify knowledge gaps and monitor learning progress effectively.

Solution

EduTraffic provides a child-friendly web platform with interactive quizzes on driving and pedestrian safety. It delivers immediate educational feedback, analyzes strengths and weaknesses, generates personalized conclusions, and provides an admin dashboard for managing questions and monitoring participant performance.



Picture 1. Score Page

Key Insights

- Interactive quizzes make traffic safety education more engaging and accessible for children.
- Category-based results reveal participants' strengths and knowledge gaps in driving and pedestrian safety.
- Immediate feedback and personalized conclusions help reinforce correct safety behavior.
- Question snapshots and local AI fallback ensure consistent and reliable assessment results.

Business / Policy Implications

EduTraffic can support schools and public institutions in delivering measurable traffic safety education.

- Supporting road-safety learning programs in schools
- Monitoring participants' understanding through an admin dashboard
- Identifying safety topics that require greater educational attention
- Providing data-driven insights for targeted awareness campaigns

This approach helps educators develop more effective and personalized traffic safety programs for children.

Tools & Technologies Used

Backend Development

- PHP 8.2+
- CodeIgniter 4.7
- Composer

Frontend Development

- HTML5
- CSS3
- JavaScript
- Responsive Web Design

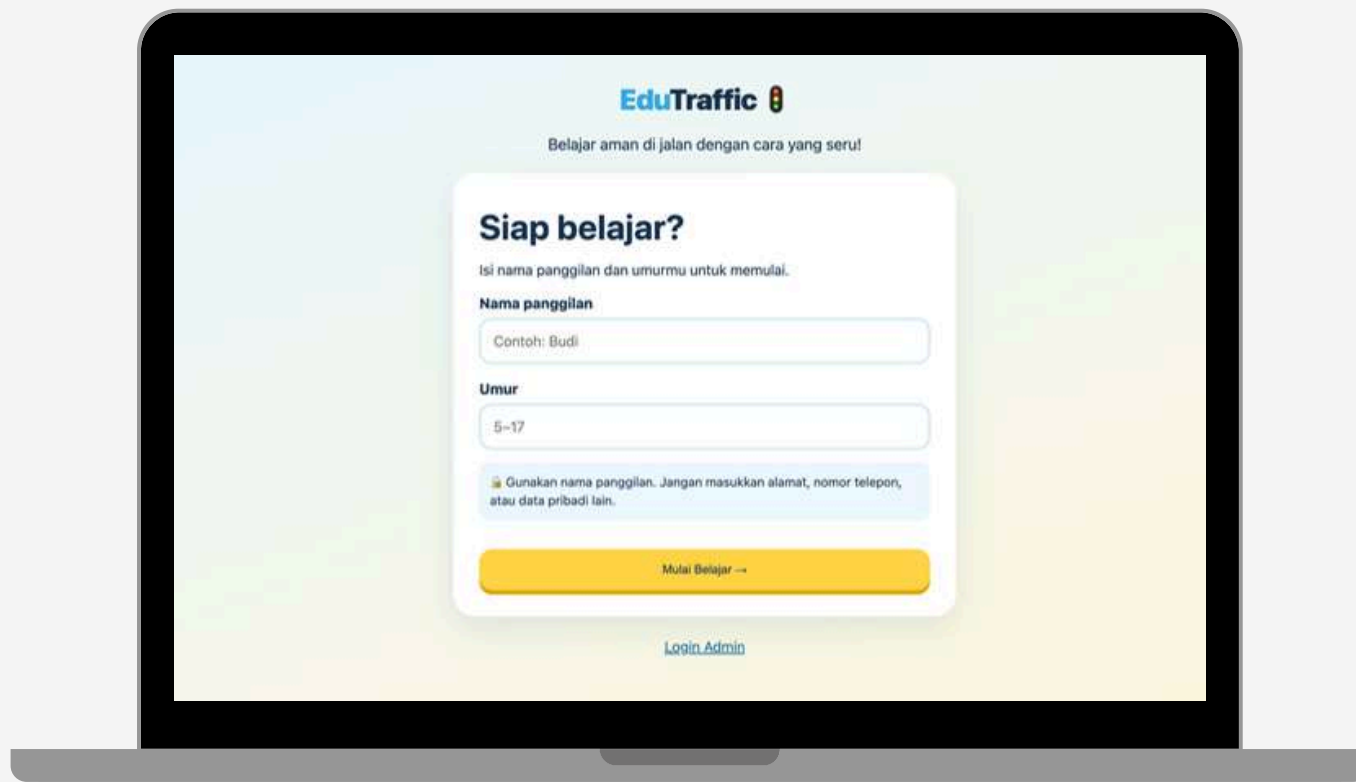
Database & Data Processing

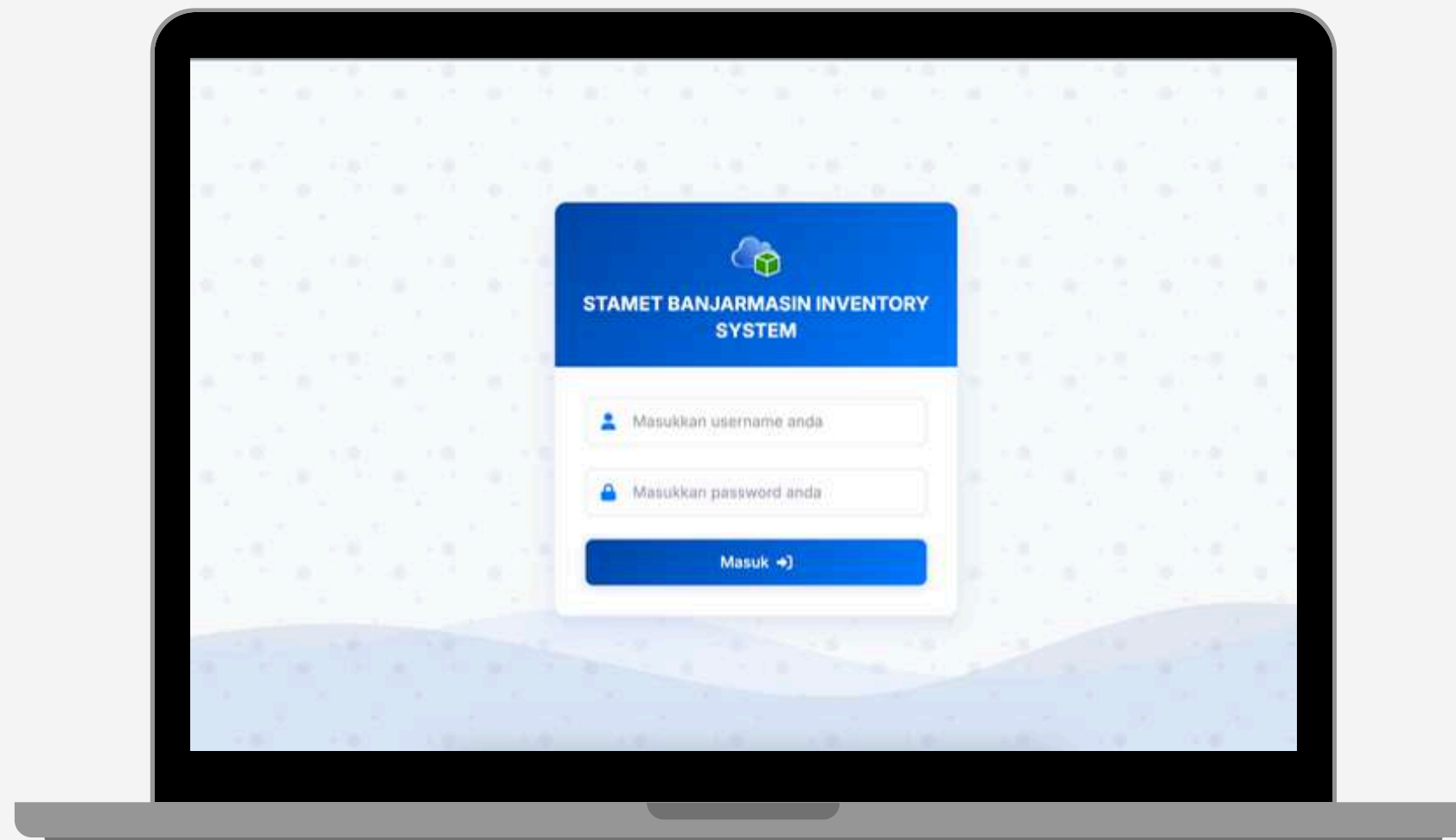
- MySQL

My Role

Full Stack Developer

Responsible for system design, backend and frontend development, database management, and API integration.





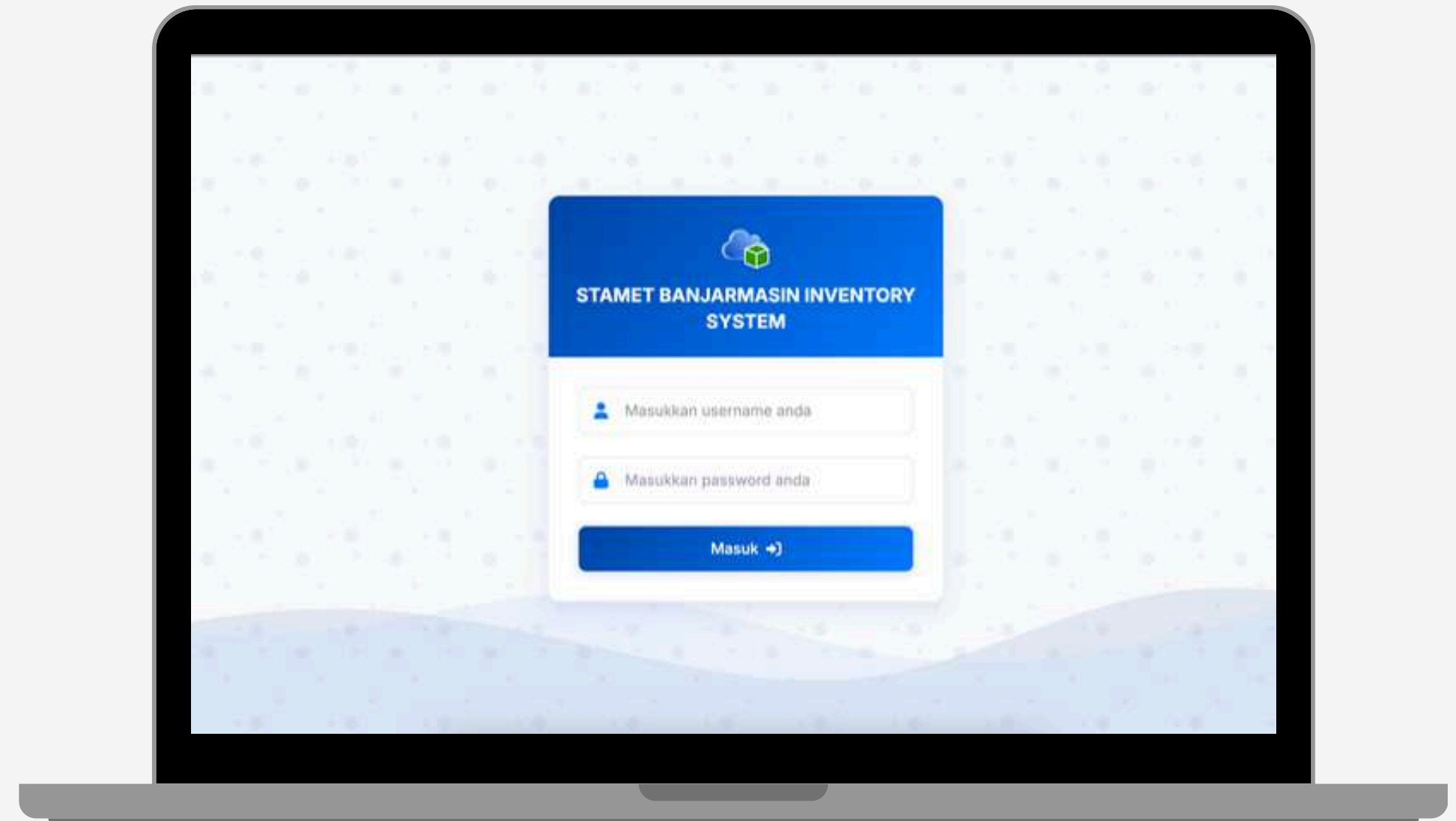
Project 3:

“Development of a Web and Mobile-Based Inventory and Asset Monitoring System for Syamsudin Noor Meteorological, Climatological, and Geophysical Agency (BMKG)”

Web Development Project

Project Overview

SiBisa (Stamet Banjarmasin Inventory System) is a digital inventory and internal loan management system developed to record, monitor, and manage inventory movement at Stasiun Meteorologi Syamsudin Noor. The system replaces manual recording with a centralized and real-time digital solution accessible via web and mobile devices.

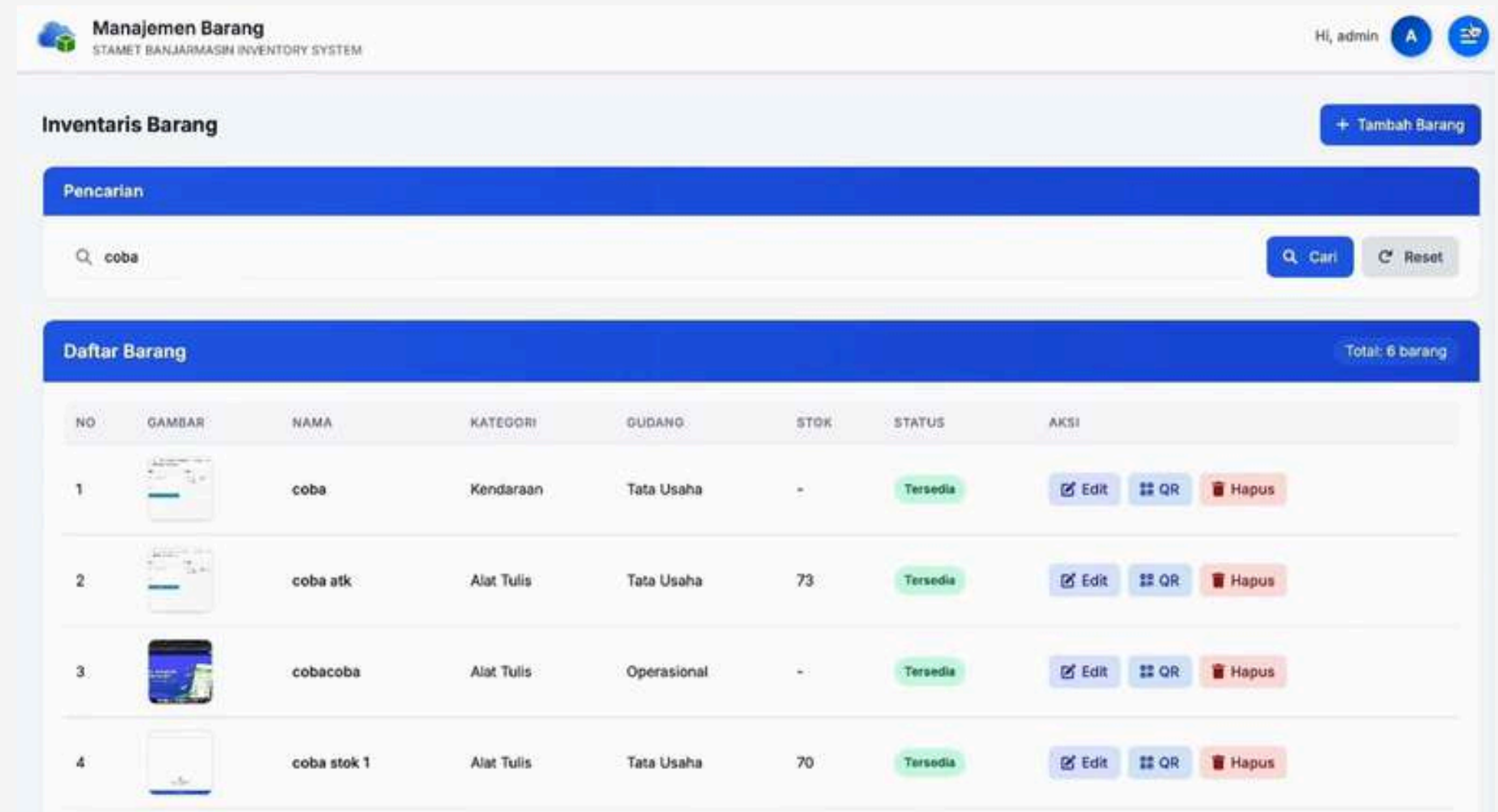


Problem

Inventory data was previously recorded manually using notebooks or separate spreadsheets, making it difficult to track item status, borrowing history, and stock availability. This caused low transparency, inefficiency, and potential data inconsistencies.

Solution

Developed a centralized system that allows authorized users to manage inventory data, track borrowing activities, and monitor item status in real time through a web admin dashboard and mobile application.





Database Structure

The system database is designed to manage inventory data, users, and borrowing transactions. It consists of three main tables: **barang**, **user**, and **riwayat_pinjam**, which are connected through relational keys.

Key Features

- User authentication & role management
- Inventory data management
- Item borrowing and return logging
- Approval workflow for borrowing requests
- Real-time item status monitoring
- QR Code scanning for faster borrowing process
- Telegram Bot notification for approvals and updates

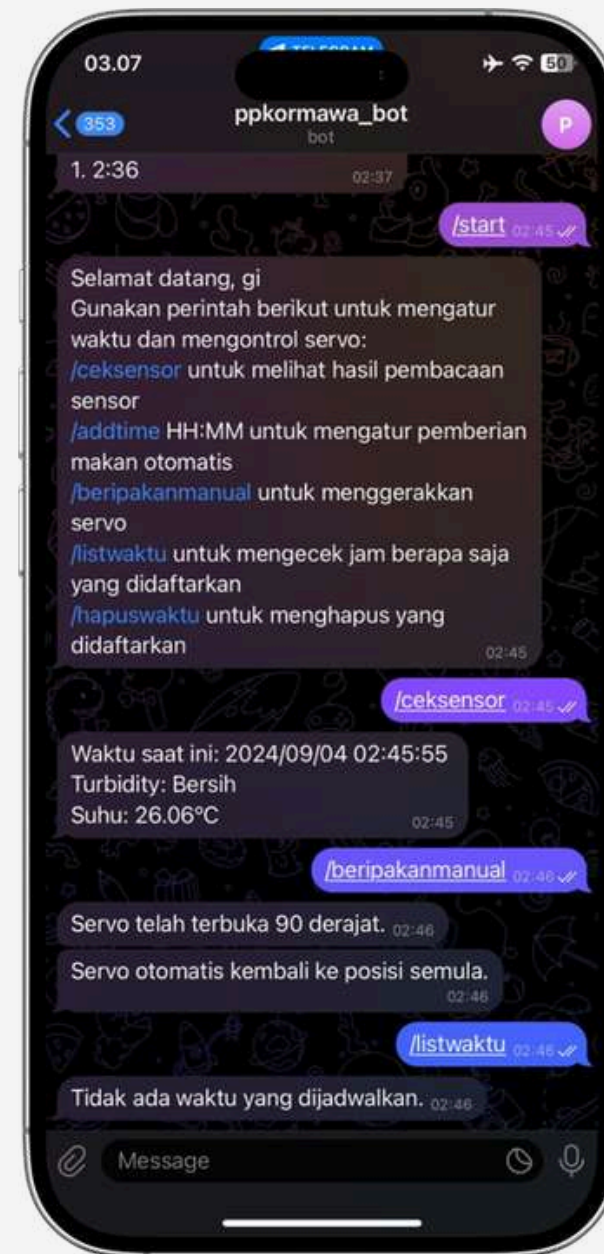
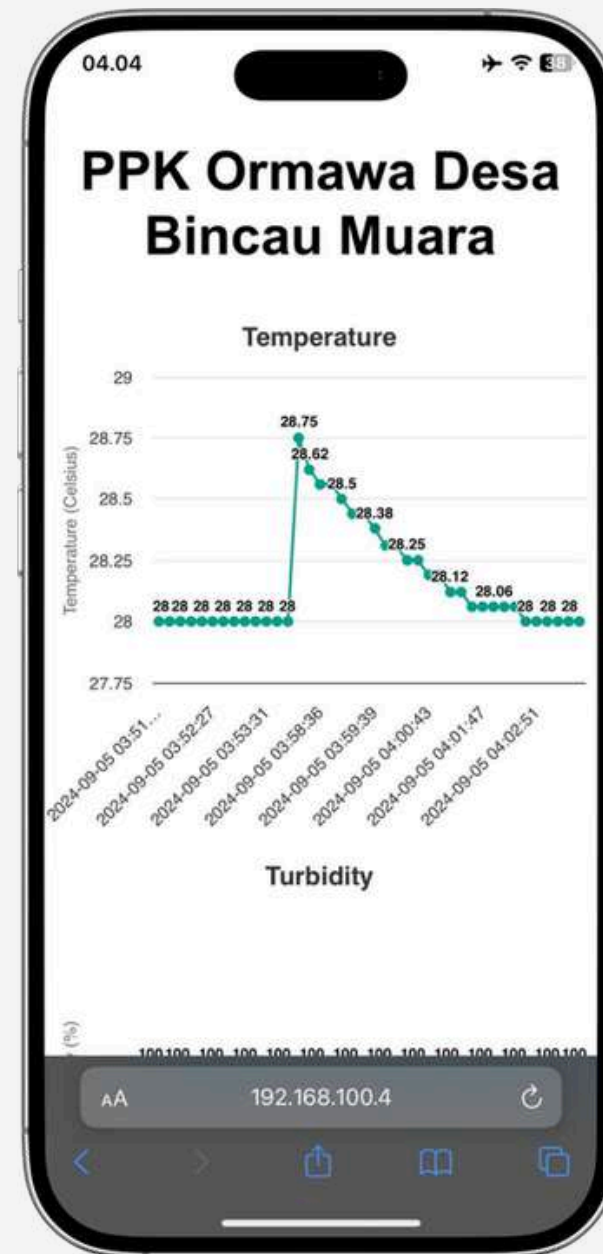
Tech Stack

- Backend: Node.js, Express.js, MySQL
- Web Admin: CodeIgniter 4, Tailwind CSS
- Mobile App: Flutter, Dart
- Integration: QR Code API, Telegram Bot

My Role

Full Stack Developer

Responsible for system design, backend and frontend development, database management, and API integration.



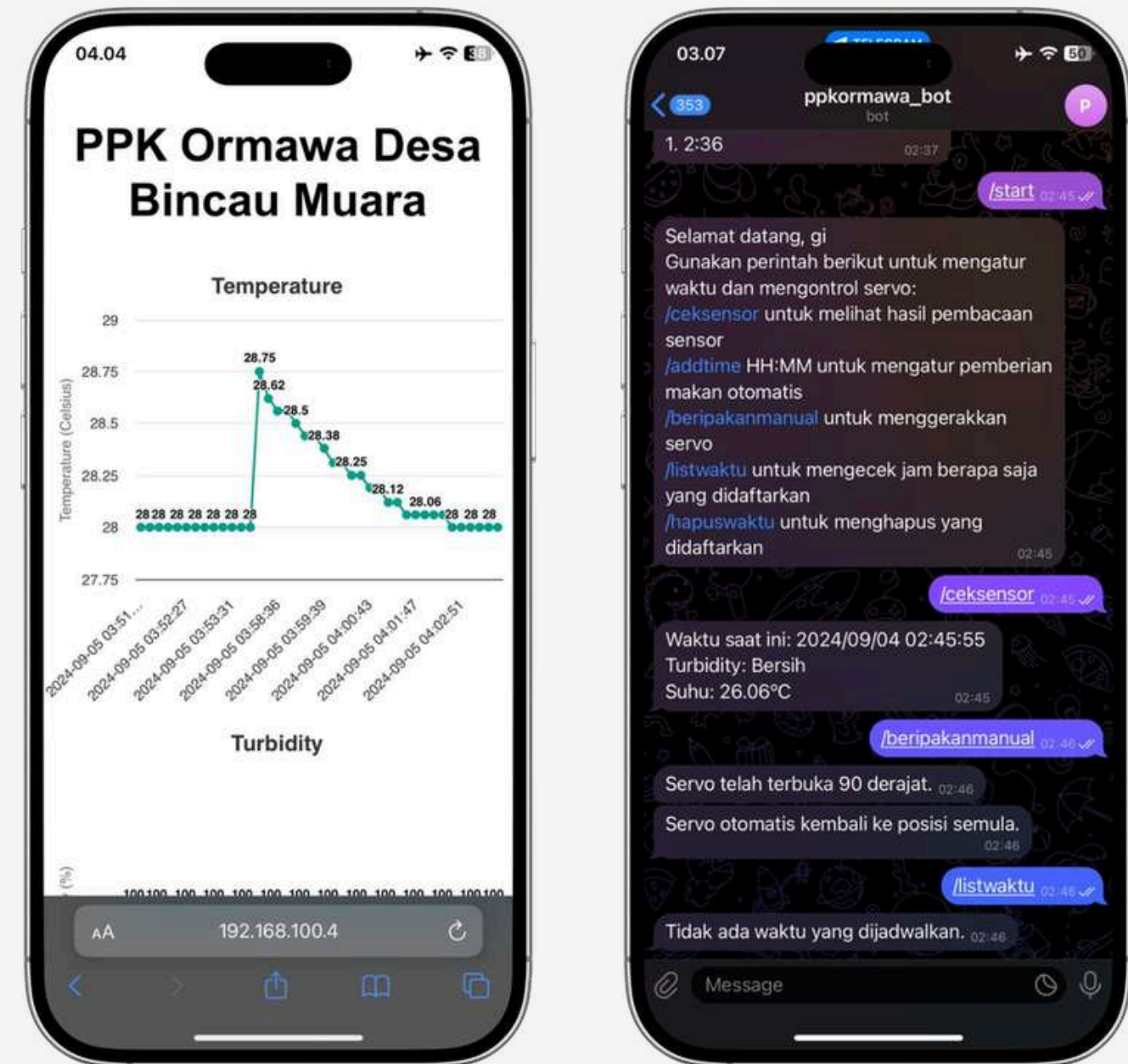
Project 4: IoT-Based Smart Fish Feeder and Water Quality Monitoring System

IoT and Web Development

Project Overview

Developed an IoT-based smart fish feeder that combines automatic fish feeding with real-time water condition monitoring. The system collects water quality data from multiple sensors, sends the data to a web-based monitoring interface, and provides Telegram notifications when specific conditions require attention.

This project helps fish owners monitor aquarium or pond conditions remotely while automating the feeding process.



Key Features

Automatic Fish Feeding – Scheduled feeding using servo motor control

Water Temperature Monitoring – Real-time temperature tracking via DS18B20

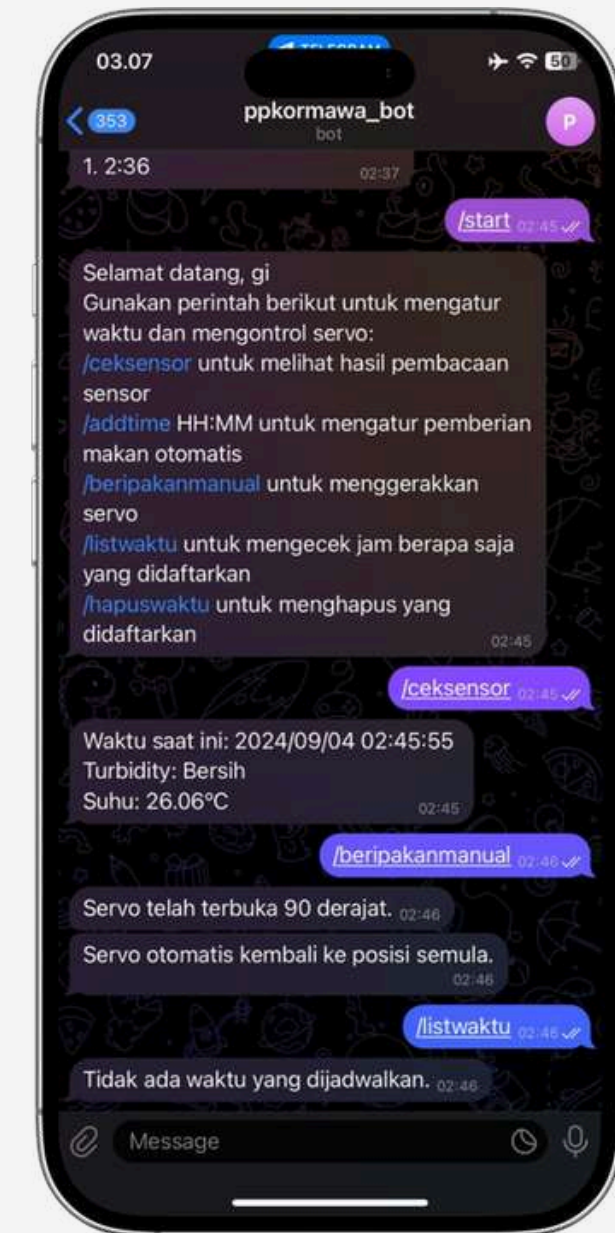
TDS Monitoring – Measures total dissolved solids in water

Water Clarity Monitoring – Turbidity sensor detects water cloudiness

Web-Based Monitoring – Live dashboard displaying all sensor data

Telegram Notification – Alerts sent when conditions need attention

Device Status Display – OLED screen shows current system status



Telegram Notification

Technology & Components

Microcontroller: ESP32

Sensors: TDS Sensor, Turbidity Sensor, DS18B20 Temperature Sensor

Actuators & Modules: Servo Motor MG90S, DC Motor, RTC DS3231, 0.96" OLED I2C

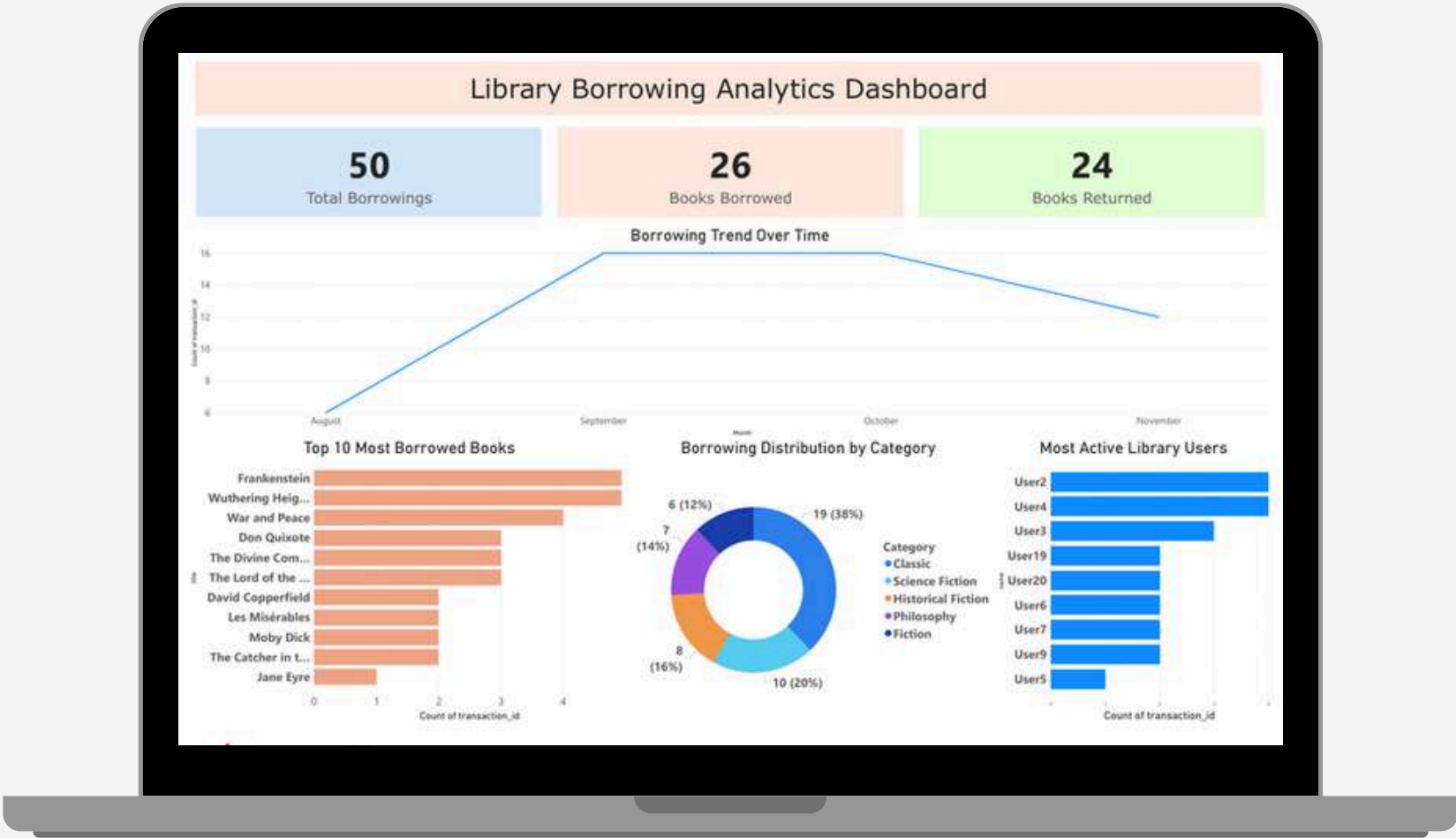
Software & Communication: IoT System Development, Web Dashboard, Telegram Bot API, Sensor Data Monitoring, Automated Device Control

My Role

Designed and developed the complete system including sensor integration, microcontroller programming, automatic feeding control, data monitoring, web integration, and Telegram notification functionality.

Project Highlights

Integrates embedded systems, sensors, automation, networking, web applications, and data monitoring into a single IoT solution. Strengthened practical experience in connecting physical devices with software-based monitoring systems.



Project 5: Library Borrowing Analytics Dashboard

PowerBI Project

Project Overview

This project analyzes borrowing activity in a library system using Power BI connected to a MySQL database.

The dashboard aims to provide insights into borrowing trends, popular books, category distribution, and user activity to help understand library usage patterns.



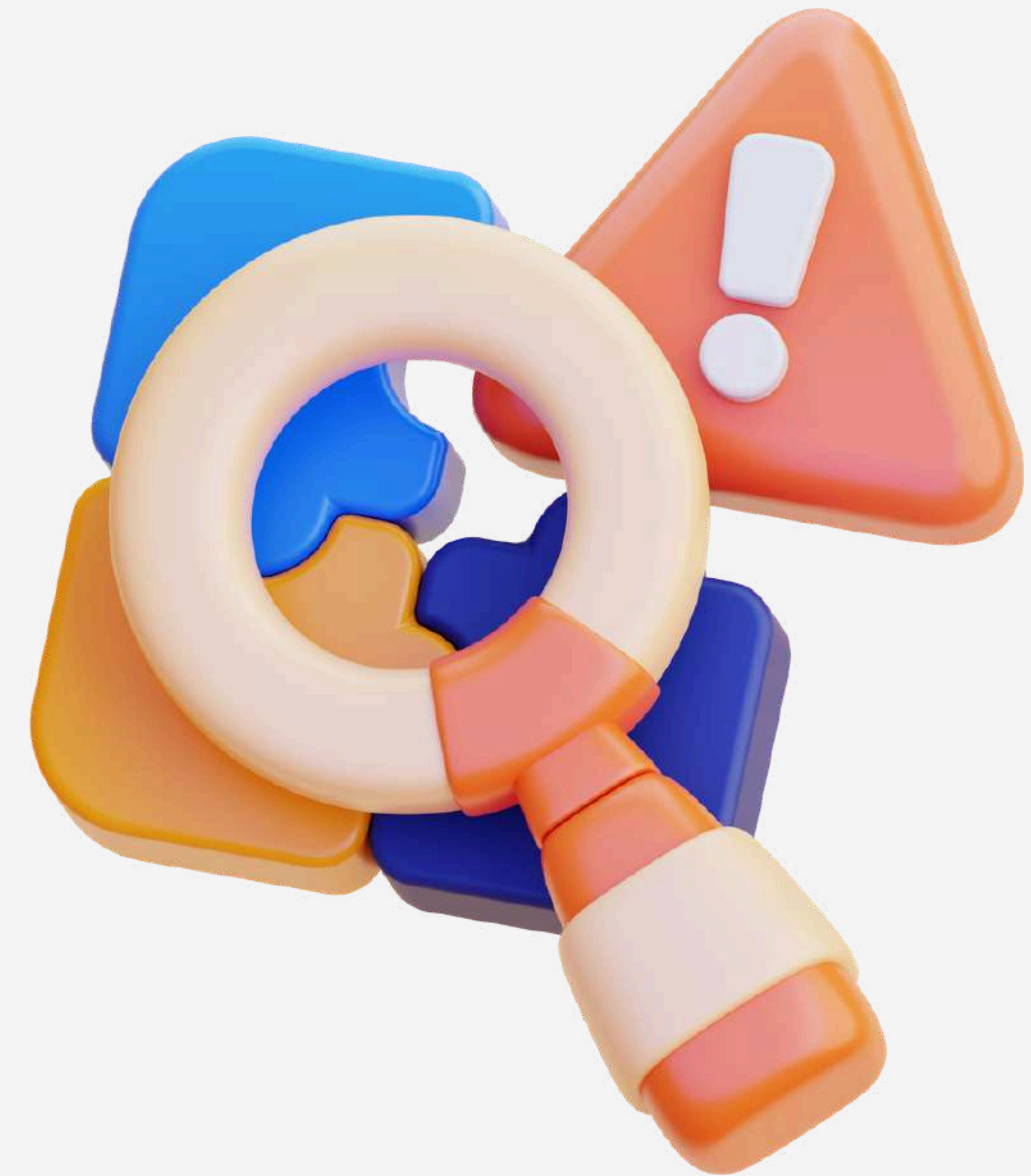
Problem Statement

Libraries need to understand borrowing behavior to manage book inventory and improve user services.

However, raw transaction data alone makes it difficult to identify trends, popular books, and active users.

This project answers several key questions:

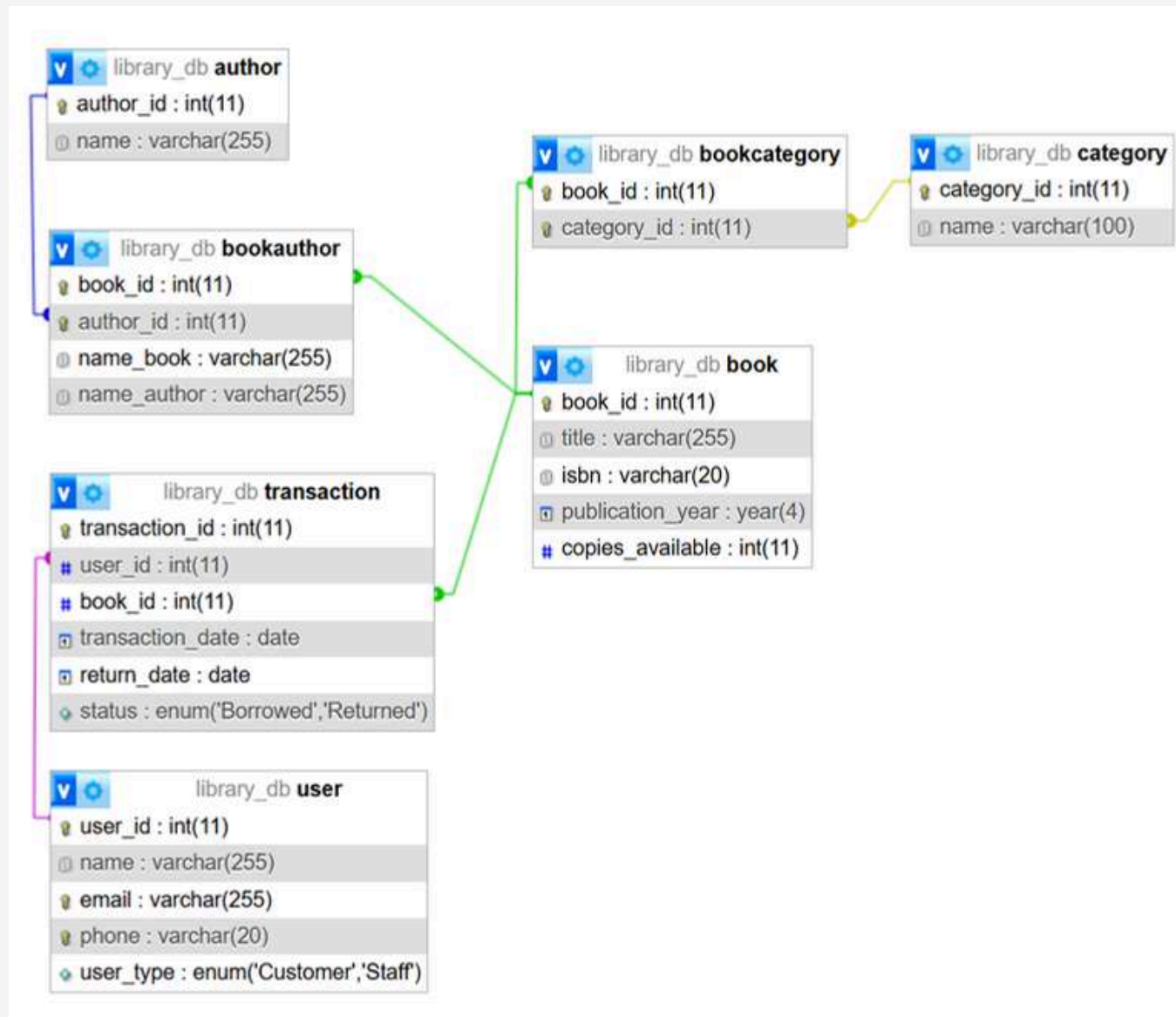
- Which books are borrowed the most?
- What categories are most popular among readers?
- How does borrowing activity change over time?
- Who are the most active library users?



Data Source

The data used in this project comes from a relational MySQL database consisting of several tables:

- book – book information (title, ISBN, publication year)
- author – author information
- category – book categories
- transaction – borrowing and return records
- user – library users
- bookauthor – relationship between books and authors
- bookcategory – relationship between books and categories





Tools & Technologies

- MySQL – database storage
- SQL – data querying
- Power BI – data visualization
- DAX – calculated measures
- ODBC Connector – connecting MySQL to Power BI

Dashboard Features

Key Performance Indicators (KPI)

- Total Borrowings
- Books Borrowed
- Books Returned

Borrowing Trend

A line chart showing how borrowing activity changes over time.

Top Borrowed Books

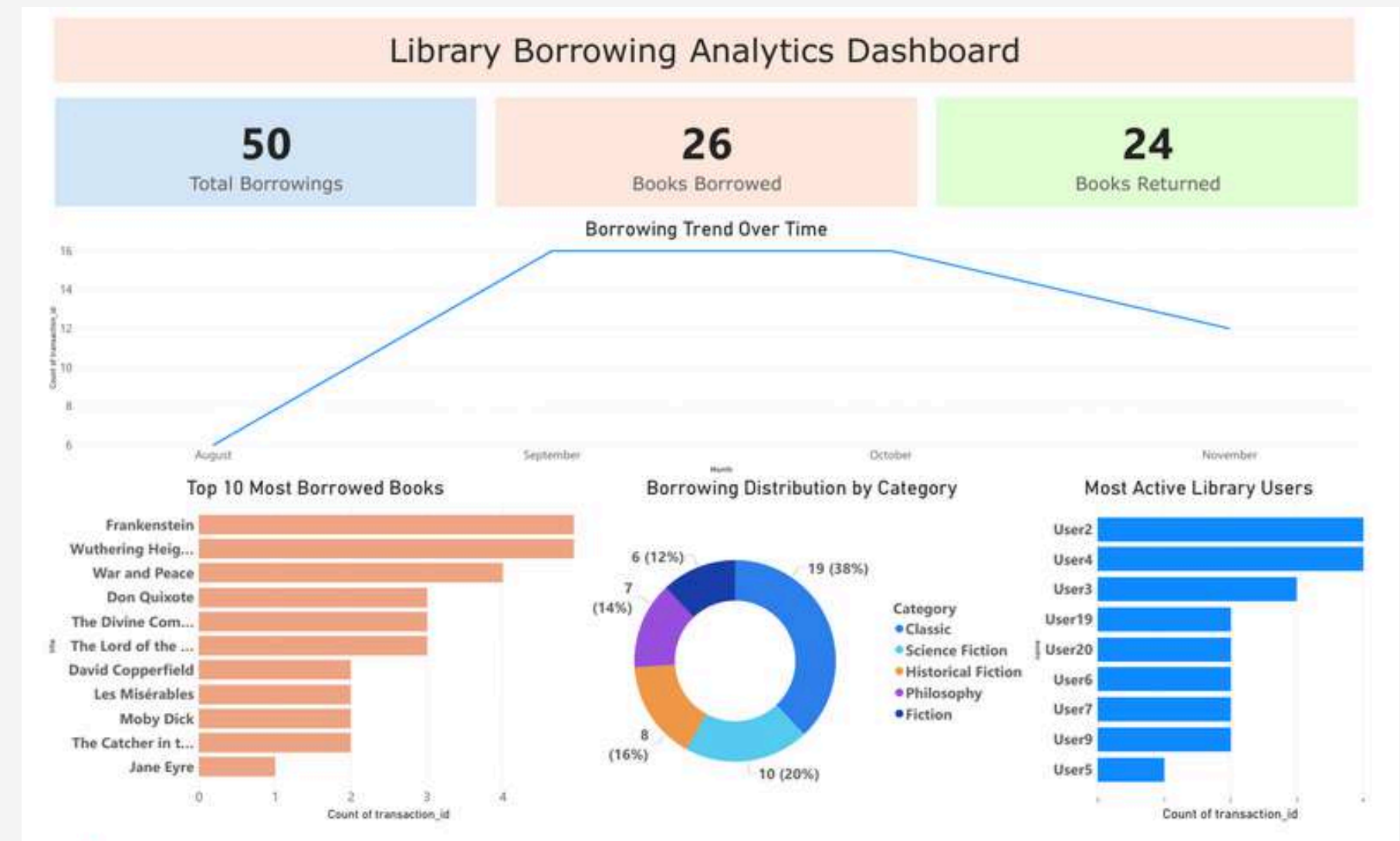
Bar chart displaying the most frequently borrowed books.

Category Distribution

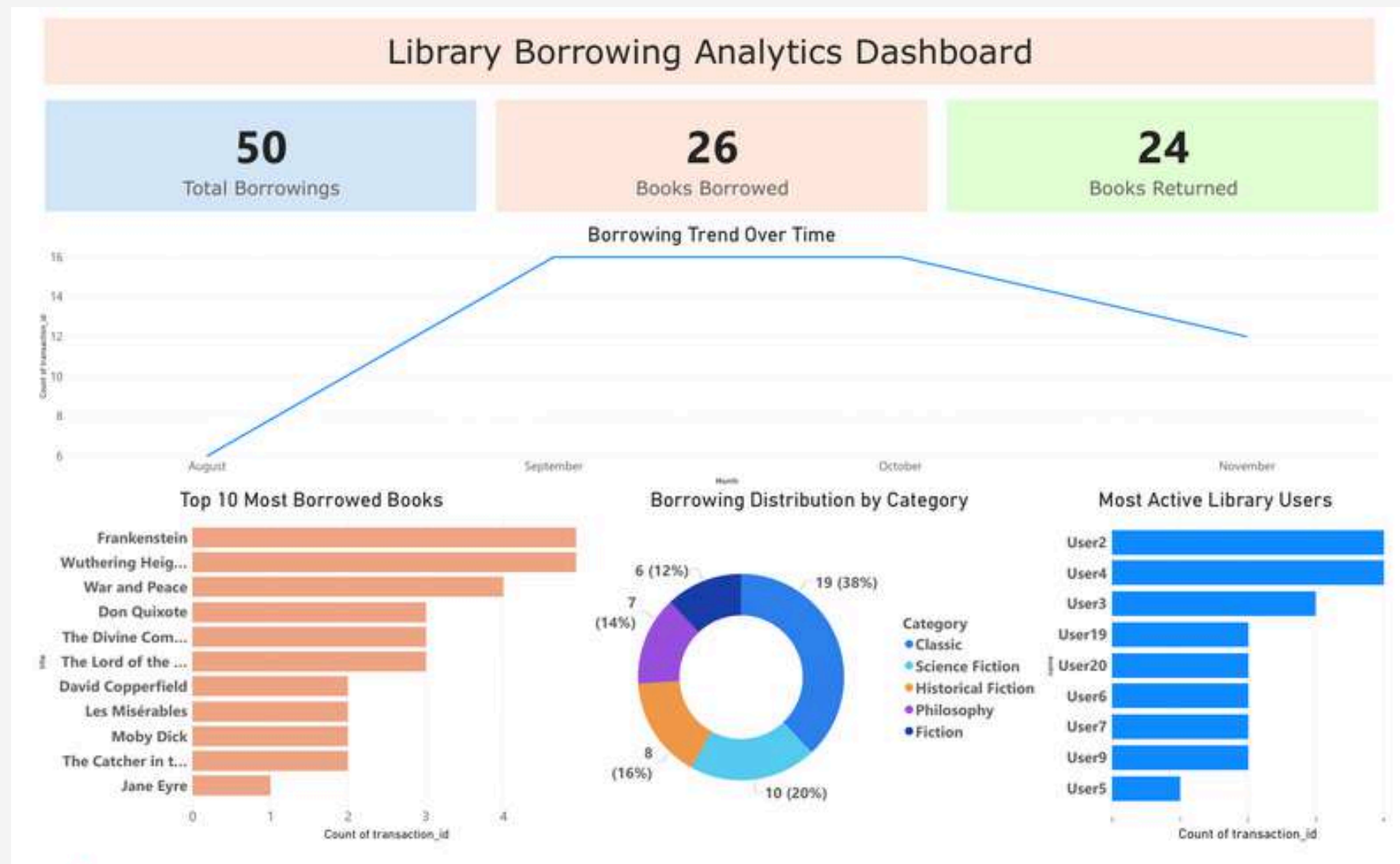
Donut chart illustrating borrowing distribution by category.

Most Active Users

Bar chart identifying the most active library members.



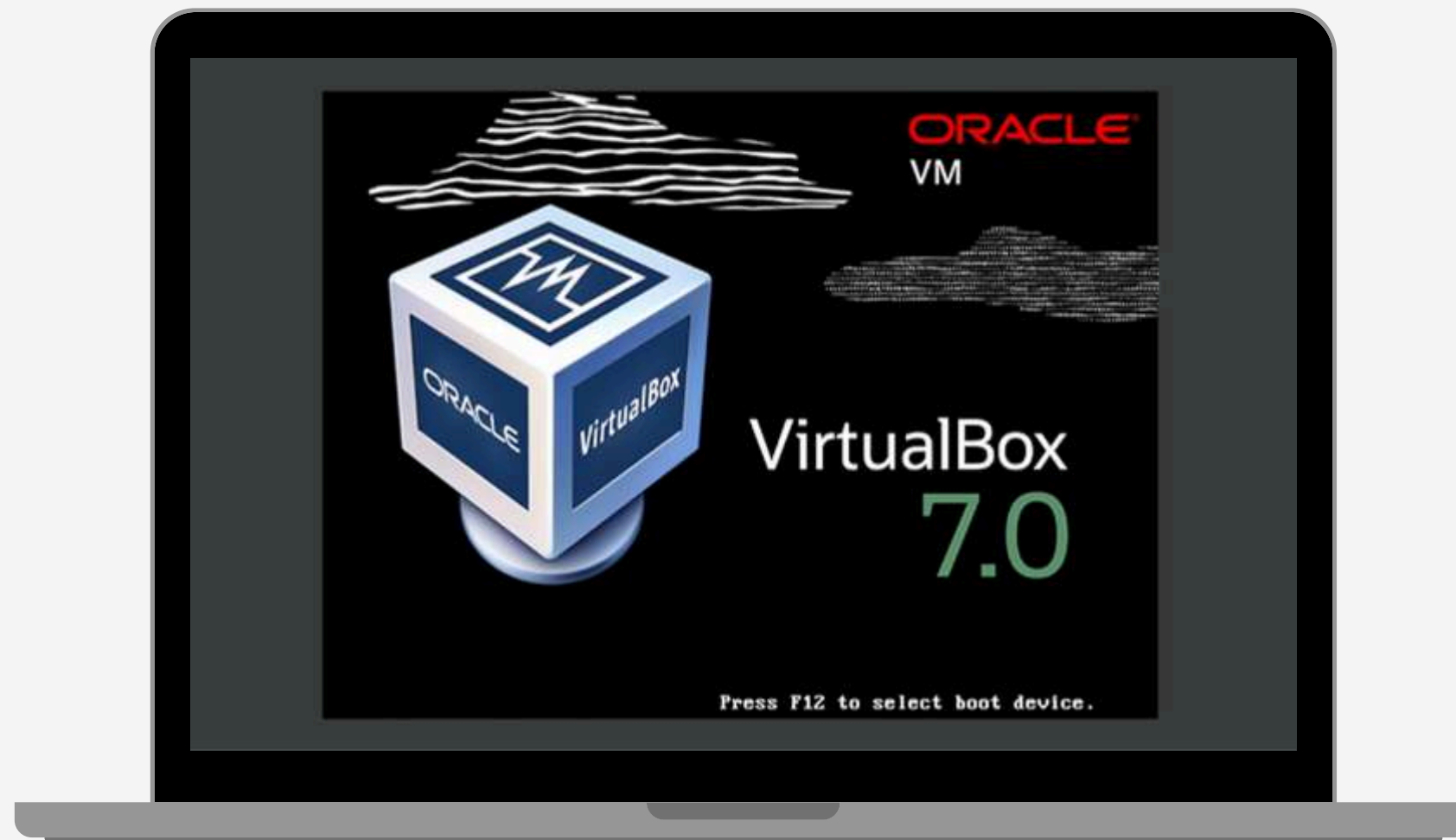
Key Insights



Several insights can be derived from the analysis:

- Classic is the most borrowed category, accounting for 38% of total transactions.
- Frankenstein and Wuthering Heights are the most popular books.
- Borrowing activity increased significantly in September compared to August.
- A small group of users contributes a higher share of borrowing activity.

These insights help identify reading preferences and user engagement patterns.



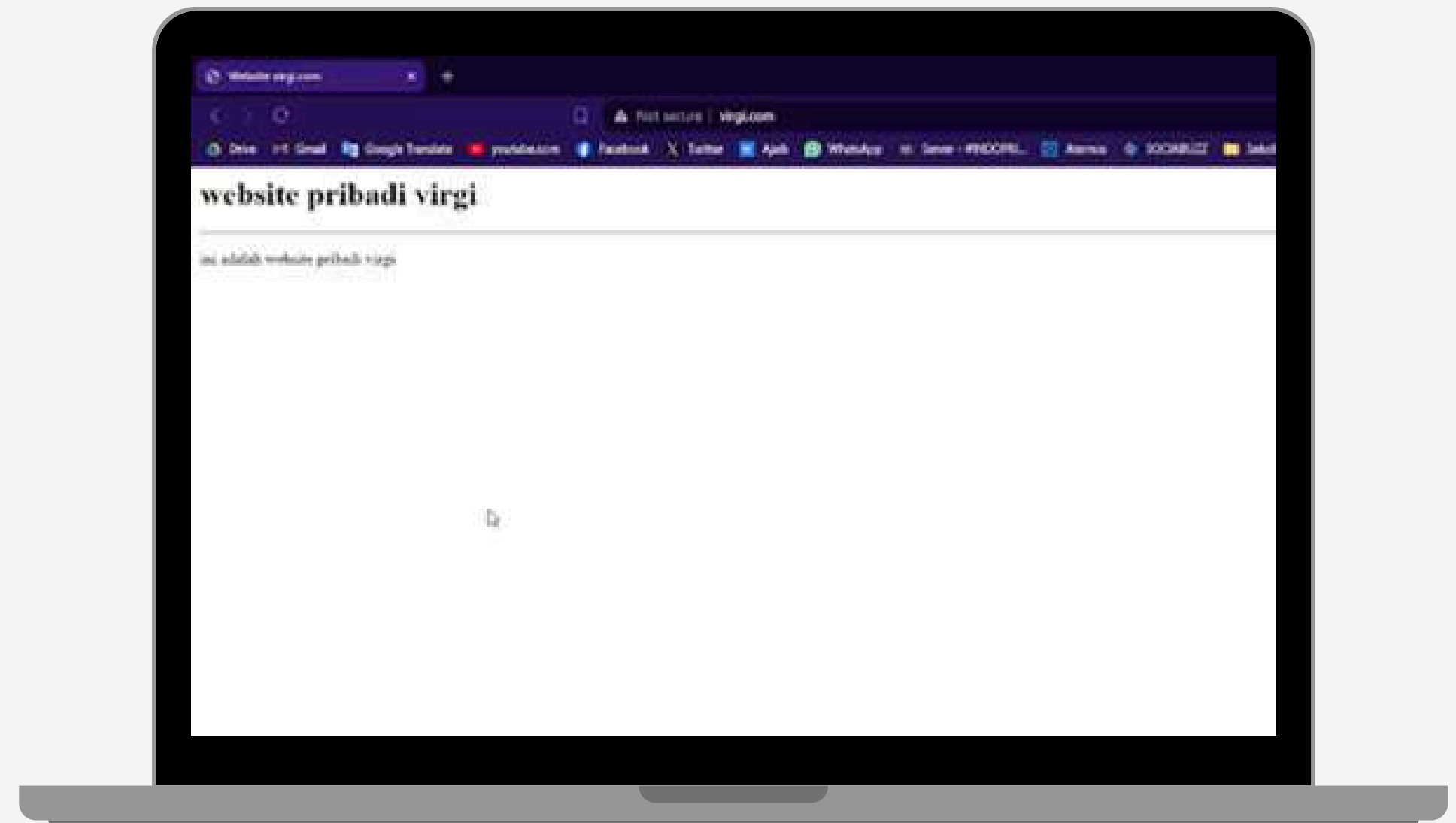
Project 6:

“Design and Implementation of a Debian 12-Based Web Server”

Network Engineering Project

Project Overview

Designed and implemented a Debian 12-based web server to provide a secure and reliable hosting environment for web applications. Configured core server services, including the web server, remote access, user management, and network settings, while applying basic security practices and performance optimization to ensure stable and efficient system operation.

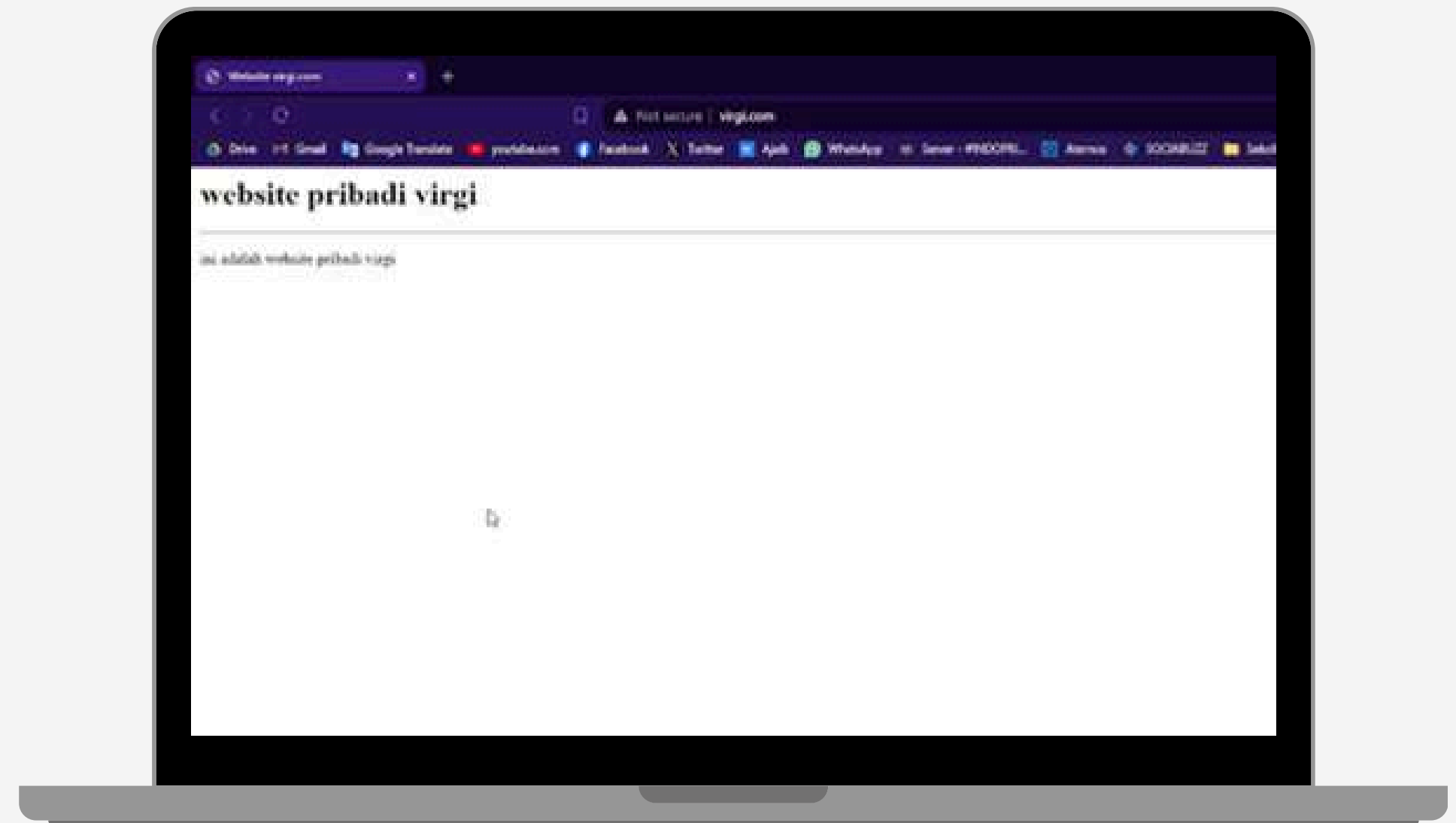


Problem

Organizations require a reliable and secure web hosting environment, but manually configuring server services and network settings can be complex, error-prone, and difficult to maintain.

Solution

Designed and deployed a Debian 12-based web server by configuring web services, user access, networking, and security settings to provide a stable, secure, and efficient hosting environment for web applications.



Key Features

- Linux server installation and configuration (Debian 12)
- Apache web server deployment
- Virtual Host configuration
- SSH remote administration
- Static IP and network configuration
- Website hosting and server monitoring

Tech Stack

- Operating System: Debian 12
- Web Server: Apache2
- Database: MySQL
- Remote Access: OpenSSH
- Networking: TCP/IP, DNS, Static IP
- Administration: Bash, Linux CLI

My Role

System Administrator & Network Engineer

Responsible for server installation, Apache configuration, network setup, and system maintenance.

Portfolio

20 Feb, 2026

Let's Discuss

Contact me:



[linkedin.com/in/virgi-atha-raditya](https://www.linkedin.com/in/virgi-atha-raditya)



+62 895 0712 6276



virgiatharaditya37@gmail.com



Banjarbaru, Kalimantan Selatan